# MD MESBAHUL MARUF

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## Education

**Ph.D. in Electrical Engineering** **Aug. 2026 – Present**
*Department of Electrical & Computer Engineering, Auburn University* *Auburn, AL*

*Focus:* Robotics, Learning-Based and Adaptive Control, Estimation, Multi-Agent Systems, and Swarm Robotics

**M.S. in Biosystems Engineering** **Jan. 2024 – July 2026**
*Auburn University* *Auburn, AL*

*Thesis: Design and Development of an Integrated In-Field Agricultural Robot for Terrain-Adaptive Mobility, Navigation, and Plant Phenotyping*.

**B.S. in Mechanical Engineering** **Jan. 2017 – Mar. 2021**
*Islamic University of Technology (IUT)* *Bangladesh*

*Thesis: Room Acoustic Parameters of an Auditorium using Ray Tracing Method*.

## Research Experience

**Continuous-Time Inverse Kalman Filtering** *2026 – Present*
*Ongoing Research Project*
- Developing a continuous-time inverse Kalman filtering framework to recover unknown process-noise covariance information from observed estimator behavior.
- Investigating model-based and data-driven (model-free) inverse estimation methods and validating recovered quantities against forward Kalman–Bucy filtering through covariance, gain, stability, convergence, and residual analyses.
- Studying identifiability, compatibility conditions, sensitivity to estimator-gain errors, and theoretical conditions for reliable inverse recovery.

**Multi-UAV Swarm Robotics and Cooperative Autonomy** *2026 – Present*
*Ongoing Research Project*
- Developing a multi-UAV swarm-robotics framework for coordinated autonomous flight, cooperative control, and multi-agent estimation using ROS2 and autopilot-based aerial platforms.
- Investigating distributed coordination, sensing, communication, localization, and experimental implementation challenges for scalable multi-agent autonomous systems.

**Autonomous Agricultural Robot for Navigation and 3D Plant Phenotyping** *2024 – 2026*
*M.S. Research*
- Designed and developed a four-wheel autonomous agricultural robot with terrain-adaptive suspension for reliable operation on uneven field terrain.
- Implemented independent PID wheel-speed control, Pure Pursuit path tracking, and EKF fusion of GPS, IMU, and wheel odometry; integrated ROS2 and multi-stereo vision for 3D plant height and volume estimation.
- Performed suspension and vehicle-dynamics analysis in MATLAB and Simscape Multibody and structural analysis in ANSYS, followed by system integration and field validation.

**System Design and Development of a Mars Rover** *2017 – 2020*
*Project Research Work – Team Leader, Team IUT Mars Rover*
- Led multidisciplinary rover development spanning mechanical design, control, computer vision, fabrication, integration, and testing for autonomous rough-terrain operation.
- Optimized the lightweight structure using SolidWorks and ANSYS and implemented GPS waypoint navigation with PID steering and velocity control plus OpenCV-based obstacle detection, terrain classification, and QR-code recognition.

**Robotic System Development – Food Delivery Robot** *2020 – 2021*
- Developed an autonomous mobile food-delivery robot integrating GPS guidance, computer vision, sensing, actuation, and mobile-robot navigation and control.

## Professional Experience

**Graduate Research Assistant – Electrical & Computer Engineering** **Aug. 2026 – Present**
*Auburn University* *Auburn, AL*

- Conducting graduate research in control, estimation, learning-based autonomous systems, and multi-agent robotics, including inverse Kalman filtering and multi-UAV and swarm-robotics research.
- Developing and evaluating algorithms through MATLAB and Simulink, ROS2-based robotic workflows, numerical simulation, stability and convergence analysis, and forward-versus-inverse validation studies.

**Graduate Research Assistant – Biosystems Engineering** **Jan. 2024 – July 2026**
*Auburn University* *Auburn, AL*

- Designed, built, and experimentally evaluated an autonomous agricultural robotic platform, including terrain-adaptive suspension, mechanical integration, sensing architecture, and field-testing workflows.
- Implemented PID control, Pure Pursuit navigation, EKF-based fusion of GPS, IMU, and wheel odometry, ROS2 communication, Simscape and ANSYS analysis, and multi-stereo 3D vision for plant phenotyping.

**Mechanical Engineer – Research & Development** **Aug. 2021 – Dec. 2022**
*Walton Group* *Gazipur, Bangladesh*

- Led CAD, FEA, GD&T, and DFM for plastic and metal production components.
- Designed injection-molded parts and improved manufacturing processes to reduce material waste and improve quality and cost efficiency.

**Robotics Engineer** **Dec. 2020 – July 2021**
*Cognition Ltd.* *Dhaka, Bangladesh*

- Designed two autonomous mobile robots covering kinematics, actuation, motion control, and sensor integration.
- Developed real-time vision tracking and obstacle avoidance and Kalman-filter-based fusion of IMU and GPS data for localization and autonomous navigation.

## Selected Engineering Projects

**GPS- and CNN-Based Autonomous Human Carrier** *2020 – 2021*

- Developed an autonomous ground vehicle using GPS waypoint guidance, CNN-based lane and path prediction, and Canny edge detection, Hough transform, and contour analysis for lane and obstacle perception.

**2-DOF Robotic Arm: Modeling and Computed-Torque Control** *2024*

- Derived forward and inverse kinematics and Euler–Lagrange dynamics and implemented computed-torque control for trajectory tracking.

**Double and Single Inverted-Pendulum Control** *2024 – 2025*

- Developed state-space and feedback-control models for stabilization and response analysis of unstable inverted-pendulum systems.

## Technical Skills

**Control & Estimation:** PID, LQR, Kalman filtering, EKF, Pure Pursuit, state-space modeling, kinematic and dynamic modeling, trajectory tracking, and optimal-control fundamentals.

**Programming & Embedded Systems:** MATLAB and Simulink, Python, C and C++, Arduino, STM32, NVIDIA Jetson, motor and sensor interfacing, and real-time robotic control.

**Robotics & Perception:** ROS2, Gazebo, CoppeliaSim, OpenCV, PyTorch, YOLOv5, CNNs, stereo vision, 3D point clouds, optical flow, and multi-sensor integration.

**CAD/CAE & Fabrication:** SolidWorks, CATIA, ANSYS, COMSOL, Abaqus, Moldflow, FEA, GD&T, DFM, 3D printing, laser cutting, and CNC machining.

## Awards & Honors

**Winner** – Farm Robotics Challenge, FIRA-USA (2024)
**Finalist** – European Rover Challenge, Poland (2020)
**2nd Runner-Up** – Indian Rover Challenge, India (2019)
**OIC Scholarship** – IUT (2017)